

PROJECT DOCUMENTATION ON

SkyCast

– Weather Forecast Application –



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INTERNSHIP DURATION

4 WEEKS

INTERNSHIP / DOMAIN

FRONTEND WEB DEVELOPMENT



INTRODUCTION

SkyCast – Weather Forecast Application is a web-based weather application designed to provide users with weather information for different locations through a simple, modern, and user-friendly interface.

The application allows users to search for a city and view important weather details such as temperature, weather conditions, humidity, wind speed, and other available information.

The main purpose of SkyCast is to make weather information easy to access and understand while demonstrating the practical use of frontend web development and API integration.

The project is developed using **HTML5, CSS3, and JavaScript**. It focuses on creating an interactive and responsive weather interface where weather data can be displayed dynamically based on the user's search.

PROJECT OBJECTIVES

The main objective of **SkyCast – Weather Forecast Application** is to develop a simple, interactive, and user-friendly web application for accessing weather information.

The major objectives of the project are:

1. To build an easy-to-use weather forecast application.
2. To allow users to search for weather information by city or location.
3. To display important weather details in a clear and organized format.
4. To dynamically update weather information based on the user's search.
5. To provide a visually attractive and responsive user interface.
6. To understand how external weather data can be integrated into a frontend web application using an **API**.
7. To gain practical experience in **HTML5, CSS3, and JavaScript**.
8. To improve knowledge of JavaScript concepts such as API requests, asynchronous operations, and dynamic webpage updates.

PROJECT SCOPE

The scope of the SkyCast project focuses on providing users with a convenient way to check weather information for different locations.

Users can enter the name of a city or location in the search field, and the application retrieves and displays the available weather information.

The application can present important weather details such as **temperature, weather condition, humidity, and wind speed**, along with suitable weather icons or visual information.

SkyCast uses weather data obtained through an external **Weather API**, allowing the application to display dynamic information instead of static data.

The application is designed with a responsive interface so that it can adapt to different screen sizes, including desktops, laptops, tablets, and supported mobile devices.

The project mainly focuses on **frontend development and API integration** for displaying weather-related information.

TECHNOLOGIES USED

The SkyCast project has been developed using the following technologies:

1. HTML5

HTML5 is used to create the basic structure of the SkyCast application.

It defines important elements such as the search bar, weather information sections, temperature display, weather details, icons, buttons, and other interface components.

2. CSS3

CSS3 is used to design and style the application.

It controls the layout, colors, typography, backgrounds, weather cards, buttons, spacing, visual effects, and responsive behavior of the website.

CSS helps make SkyCast visually attractive and user-friendly.

3. JavaScript

JavaScript provides the main functionality and interactivity of the application.

It handles user searches, API requests, weather data processing, error handling, and dynamic updates to the webpage.

4. Weather API

A Weather API is used to retrieve weather information for the location searched by the user.

The API provides weather-related data that JavaScript processes and displays on the application interface.

5. Font Awesome

Font Awesome can be used to provide icons for interface elements such as search, location, humidity, wind, and other weather-related information.

6. Google Fonts

Google Fonts is used to provide modern and attractive typography, improving the overall appearance and readability of the application.

FEATURES

SkyCast provides several useful features for accessing and viewing weather information.

1. City Search

Users can enter the name of a city or location to search for its weather information.

2. Weather Information

The application displays important weather information in a clear and organized format.

3. Temperature Display

The current temperature of the searched location is displayed prominently so that users can quickly understand the weather condition.

4. Weather Condition

The application displays the current weather condition, such as clear, cloudy, rainy, or other available conditions based on the weather data received.

5. Humidity Information

Humidity information is displayed to provide users with additional details about atmospheric moisture.

6. Wind Speed

The application displays available wind speed information for the searched location.

7. Weather Icons

Suitable weather icons or images are displayed according to the current weather condition, making the interface easier to understand visually.

8. Dynamic Weather Data

Weather information is dynamically updated whenever the user searches for a different location.

9. API Integration

The application integrates with a Weather API to retrieve and display weather information dynamically.

10. Error Handling

The application can handle invalid or unavailable city searches by displaying an appropriate message instead of incorrect weather information.

11. Responsive Design

SkyCast is designed to adjust to different screen sizes, providing a convenient experience across desktops, laptops, tablets, and supported mobile devices.

12. User-Friendly Interface

The application provides a clean, simple, and visually attractive interface that makes weather information easy to access and understand.

PROJECT STRUCTURE

The SkyCast project files are organized in a simple and structured manner as follows:

SkyCast/

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└─ index.html

└─ style.css

└─ script.js

└─ README.md

└─ screenshots/

└─ Documentation.pdf

File Description

index.html – Contains the main HTML structure of the SkyCast application, including the search area, weather display sections, and other interface elements.

style.css – Contains the styling of the application, including layouts, backgrounds, typography, weather cards, responsive design, and visual effects.

script.js – Contains the JavaScript logic responsible for searching locations, connecting with the Weather API, retrieving weather data, and dynamically displaying information.

README.md – Contains the project overview, features, technologies used, project information, and screenshots.

- **screenshots/** – Contains screenshots of the actual SkyCast application

Documentation.pdf – Contains the complete documentation of the SkyCast project.

WORKING OF THE PROJECT

The basic working flow of SkyCast can be represented as:

Open SkyCast → Enter City → Search → API Request → Receive Weather Data → Display Weather Information

Step 1: Open the Application

The user opens the SkyCast application and is presented with the main weather interface.

The application provides a search option where the user can enter a city or location.

Step 2: Enter a City

The user enters the name of the city for which weather information is required.

For example, the user can search for a city such as **Bhubaneswar, Kolkata, Delhi, or Mumbai**.

Step 3: Search for Weather

After entering the city name, the user clicks the search button or performs the available search action.

JavaScript reads the entered city name and prepares a request for weather information.

Step 4: API Request

The application sends a request to the connected Weather API.

The requested city or location is included so that the API can identify the correct weather information.

Step 5: Receive Weather Data

The Weather API processes the request and returns available weather data for the searched location.

JavaScript receives and processes this information.

Step 6: Display Weather Information

The application dynamically displays the available weather details on the screen.

These may include:

- City or location name
- Temperature
- Weather condition
- Humidity
- Wind speed
- Weather icon

The displayed information changes dynamically whenever the user searches for another city.

Step 7: Invalid Search Handling

If the user enters an invalid or unavailable city name, the application handles the error and displays an appropriate message.

This prevents incorrect or unavailable data from being shown to the user.

CONCLUSION

The **SkyCast – Weather Forecast Application** project was successfully developed as a frontend web application for accessing and displaying weather information through an interactive and user-friendly interface.

The project demonstrates important functionalities such as **city-based weather search, dynamic weather information, temperature display, weather conditions, humidity, wind speed, weather icons, API integration, and responsive design.**

Through the development of SkyCast, I gained practical experience in using **HTML5, CSS3, and JavaScript** to build a functional web application. I also improved my understanding of **API integration, asynchronous JavaScript, data retrieval, dynamic webpage updates, and error handling.**

This project provided valuable practical experience in understanding how a frontend application can communicate with an external API and use the received data to provide useful information to users.

Overall, the SkyCast project strengthened my frontend web development skills and enhanced my knowledge of developing **interactive, responsive, and API-based web applications.**